Computer Graphics and Man-Machine Interface

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CMPn322 - CMPN322 - CMP 422

Geometric transformation functions

Translation





Scaling





Rotation



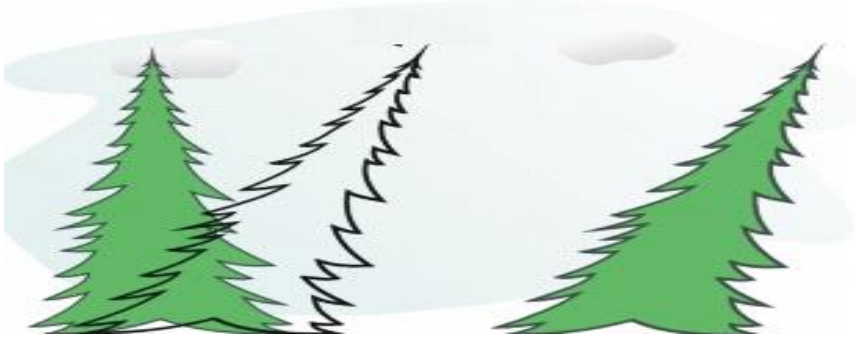


Reflection





Shearing

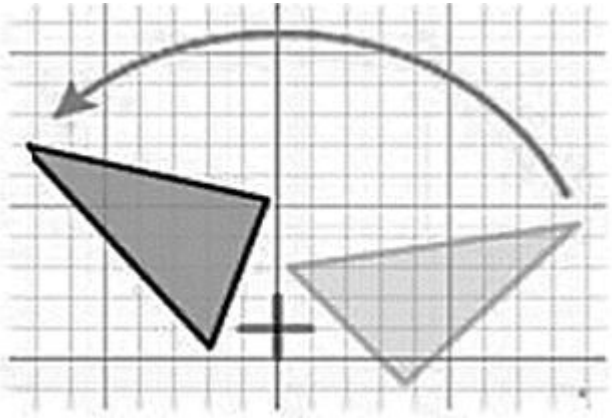


Geometric Transformation

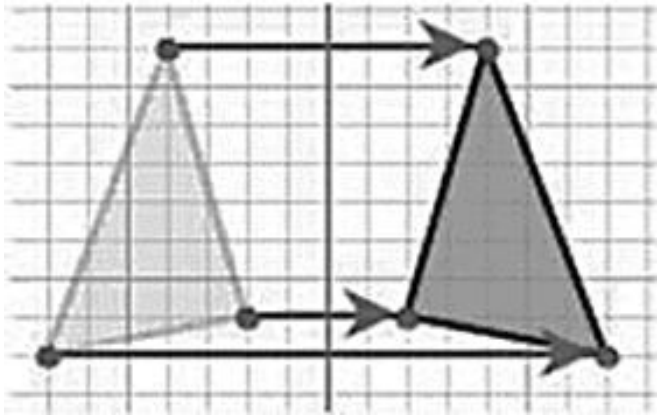
- Geometric transformations map points in one space to points in another.
- These transformations can **linear** or **non-linear** transformations
- We'll focus on **linear transformations** that preserves co-linearity and ratios of distances (parallel lines remain parallel)
  - Maps origin to origin
  - Co-linearity→ all points lying on a line initially still lie on a line after transformation
  - Ratios of distances→ the midpoint of a line segment remains the midpoint after transformation .

Linear Transformation

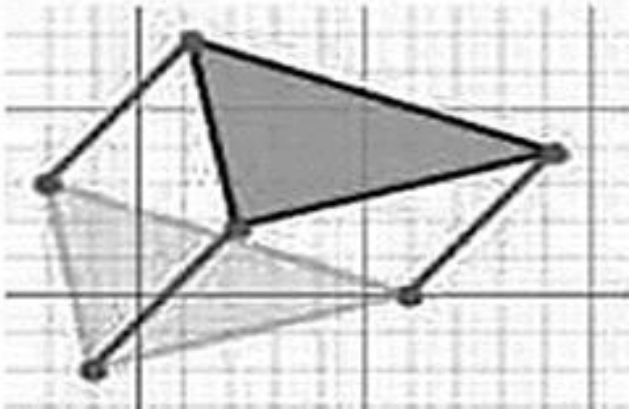
- Linear transformations
  - After transformation the distance ratio remain constant, and the parallel lines stay parallel



Rotation



Reflection



Translation

Linear Transformation

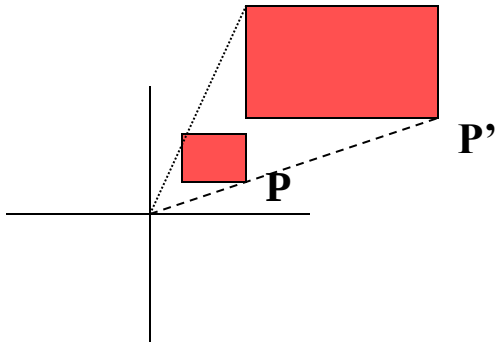
- Linear Transformation can be represented by(non-singular) matrices:
  - $\begin{bmatrix} x' \\ y' \end{bmatrix} = M \begin{bmatrix} x \\ y \end{bmatrix}$
    - M a matrix representing linear transformation which has to be square matrix of n columns and n rows.
    - X', y' represent the new position of x and y after transformation
      - $M = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$
      - $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$ → x'=ax+by &y'=cx+dy

2D Scaling from the origin

Point P defined as $P(x, y)$,

Perform a scale (stretch) to Point $P'(x', y')$ by a factor s_x along the x axis, and s_y along the y axis.

$x' = s_x . x, \qquad y' = s_y . y$



In matrix form

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix} . \begin{bmatrix} x \\ y \end{bmatrix}$$

Define matrix $S = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix}$,

Properties of scaling :

- Doesn't preserve angles between lines except with uniform scaling sx=sy.
- The object must be centered on origin, or it will be shifted

2D Reflection

- **Reflection creates a mirror image of an object across a line or point**
- Perform a horizontal reflection to Point $P'(x', y')$ is coresponding to a negative factor $s_y = -1$

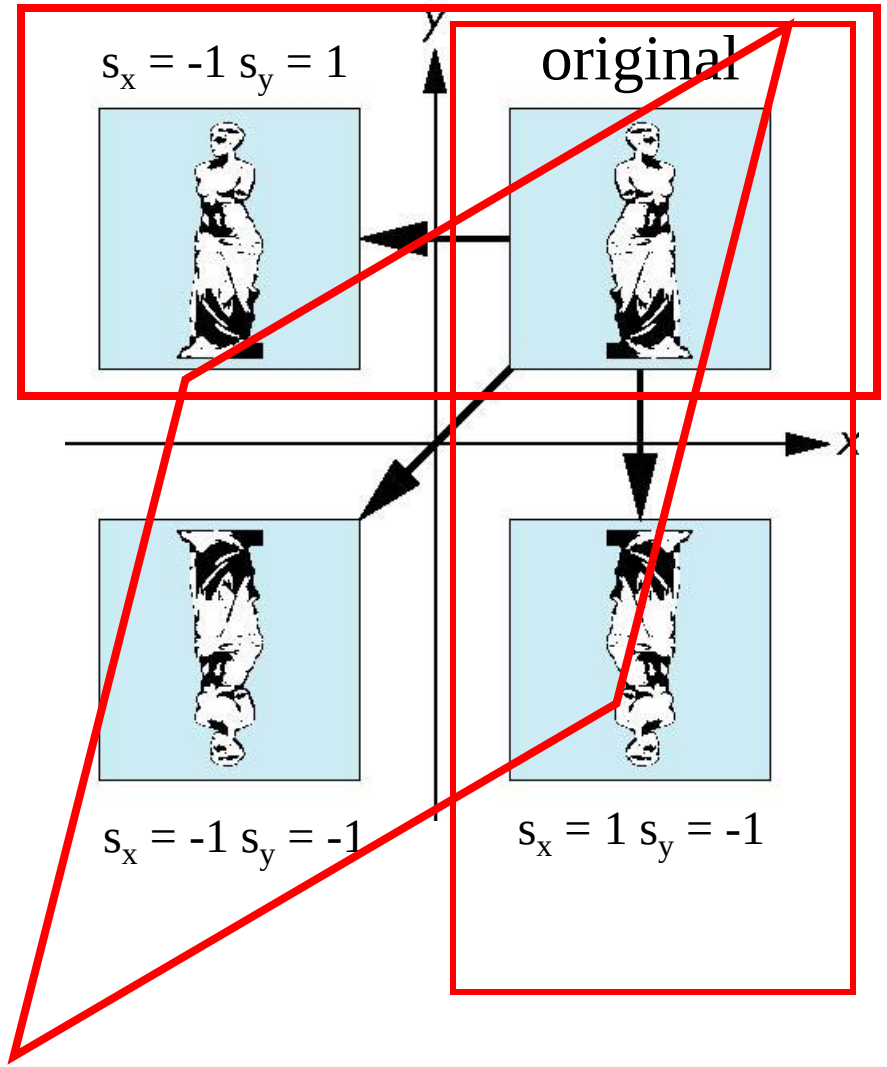
In matrix form $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} . \begin{bmatrix} x \\ y \end{bmatrix} \rightarrow x' = x, \quad y' = -y$

- Perform avertical reflection to Point $P'(x', y')$ is coresponding to a negative factor $s_x = -1$

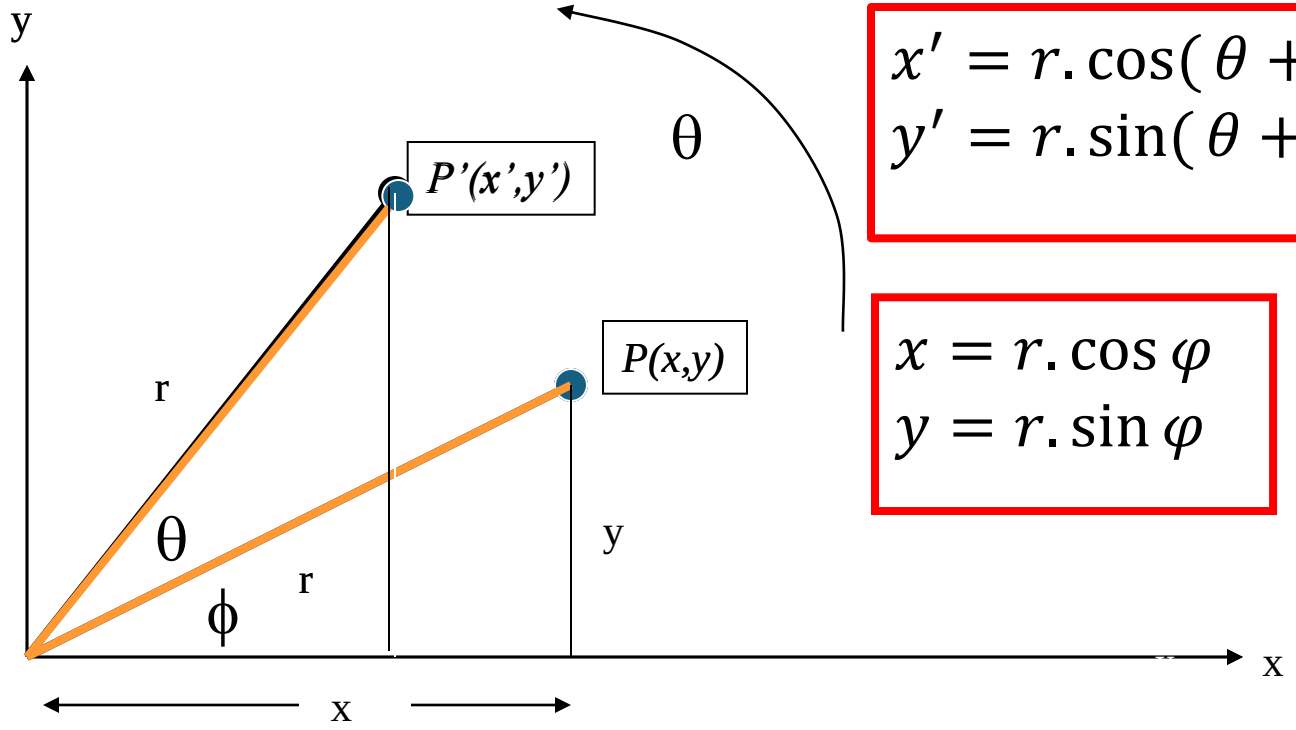
In matrix form $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} . \begin{bmatrix} x \\ y \end{bmatrix} \rightarrow x' = -x, \quad y' = y$

- Perform diagonal reflection to Point $P'(x', y')$ is coresponding to a negative factor $s_x = -1$ and negative factor $s_y = -1$

In matrix form $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} . \begin{bmatrix} x \\ y \end{bmatrix} \rightarrow x' = -x, \quad y' = -y$



2D Rotation about the origin



$$x' = r.\cos(\theta + \varphi) = r.\cos\theta.\cos\varphi - r.\sin\theta.\sin\varphi$$
$$y' = r.\sin(\theta + \varphi) = r.\sin\theta.\cos\varphi + r.\cos\theta.\sin\varphi$$

$$x = r.\cos\varphi$$
$$y = r.\sin\varphi$$

2D Rotation about the origin

$$x' = r.\cos(\theta + \varphi) = r.\cos\varphi\,.\cos\theta - r.\sin\varphi\,.\sin\theta$$
$$y' = r.\sin(\theta + \varphi) = r.\cos\varphi\,.\sin\theta + r.\sin\varphi\,.\cos\theta$$

Substituting for $x = r.\cos\varphi$, $y = r.\sin\varphi$

Gives us :

$$x' = x.\cos\theta - y.\sin\theta$$
$$y' = x.\sin\theta + y.\cos\theta$$

2D Rotation about the origin

$$x' = x.\cos\theta - y.\sin\theta$$
$$y' = x.\sin\theta + y.\cos\theta$$

In matrix form gives us :

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} . \begin{bmatrix} x \\ y \end{bmatrix}$$

Define the matrix $R = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$, $\qquad P' = R \cdot P$

Shear or skewing

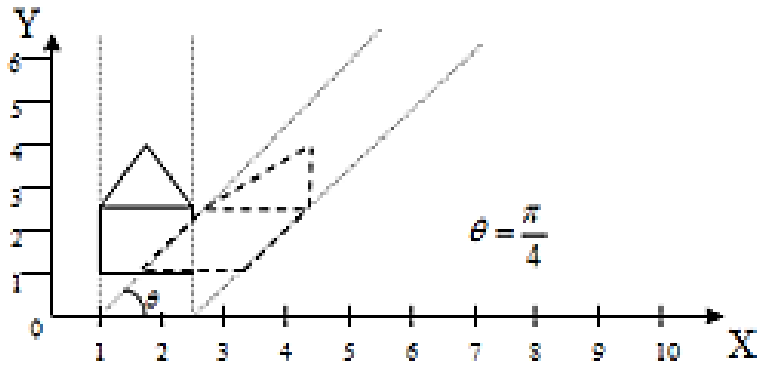
Shrear: Deform shape like shifted slices#

shear in x: $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & sh_x \\ 0 & 1 \end{bmatrix} . \begin{bmatrix} x \\ y \end{bmatrix}$

$x' = x + sh_x . y\,, \quad y' = y$

shear in y: $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ sh_y & 1 \end{bmatrix} . \begin{bmatrix} x \\ y \end{bmatrix}$

$x' = x\,, \qquad y' = sh_y . x + y$



$Skew_\theta = \begin{bmatrix} 1 & \frac{1}{\tan\theta} \\ 0 & 1 \end{bmatrix}$

2D Translation

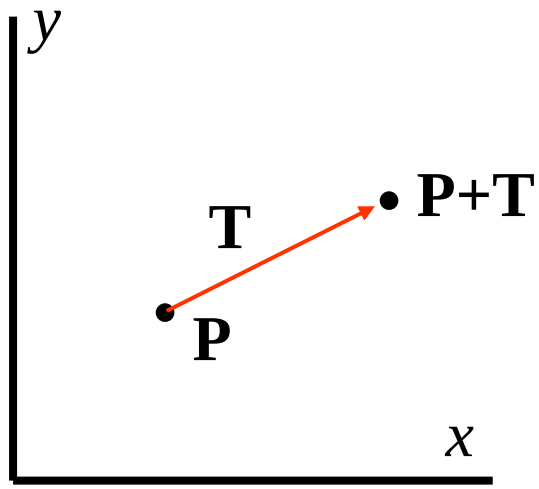
Translate over vector (t_x, t_y)

$$x' = x + t_x, \quad y' = y + t_y$$

or

$$\mathbf{P}' = \mathbf{P} + \mathbf{T}, \text{ with}$$

$$\mathbf{P}' = \begin{pmatrix} x' \\ y' \end{pmatrix}, \mathbf{P} = \begin{pmatrix} x \\ y \end{pmatrix} \text{ and } \mathbf{T} = \begin{pmatrix} t_x \\ t_y \end{pmatrix}$$



Transformations

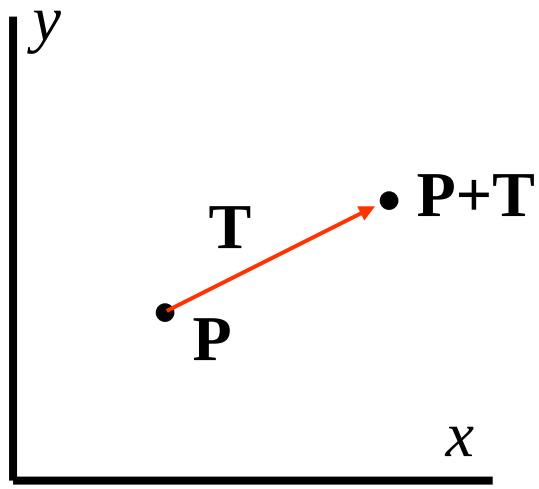
- Scale
  - $P' = S \cdot P$
- Rotation
  - $P' = R \cdot P$
- shear
  - $P' = shr \cdot P$
- Translation
  - $P' = T + P$

- **We would like all transformations to be multiplications so we can concatenate them to combine transformations**

2D Translation

Translate over vector (t_x, t_y)

$$x' = x + t_x, \quad y' = y + t_y$$



- In matrix form
- $$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \end{bmatrix} \cdot \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Define the matrix $T = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \end{bmatrix}$,

$$P' = T \cdot P$$

Homogeneous coordinates

- Uniform representation of translation, rotation, scaling
- Uniform representation of points and vectors
- Compact representation of a sequence of transformations

To make a uniform representation to all transformation matrices

- Add an extra coordinate, W (weight=1), to a point.
  - P(x, y)➔ P(x, y,1).

Translations in homogenised coordinates

- Transformation matrices for 2D translation are now 3x3.

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \qquad \begin{aligned} x' &= x + t_x \\ y' &= y + t_y \\ 1 &= 1 \end{aligned}$$

- Inverse translation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -t_x \\ 0 & 1 & -t_y \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \qquad \begin{aligned} x' &= x - t_x \\ y' &= y - t_y \\ 1 &= 1 \end{aligned}$$

Homogeneous form of Shear or skewing

Recall the (x,y) form of shear in x: $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & sh_x \\ 0 & 1 \end{bmatrix} . \begin{bmatrix} x \\ y \end{bmatrix}$

Recall the (x,y) form of shear in y: $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ sh_y & 1 \end{bmatrix} . \begin{bmatrix} x \\ y \end{bmatrix}$

In homogeneous coordinates :

shear in x: $\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & sh_x & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \rightarrow x' = x + sh_x . y \, , \quad y' = y$

shear in y: $\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ sh_y & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} x \\ y \\ 1 \end{bmatrix} \rightarrow x' = x \, , \quad y' = sh_y . x + y,$

Homogeneous form of scale

Recall the (x,y) form of Scale :

$$S(s_x, s_y) = \begin{bmatrix} s_x & 0 \\ 0 & s_y \end{bmatrix}$$

In homogeneous coordinates :

$$S(s_x, s_y) = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Inverse Scaling: $\mathbf{S}^{-1}(s_x, s_y)$:

$$\mathbf{S}^{-1}(s_x, s_y) = \begin{bmatrix} 1/s_x & 0 & 0 \\ 0 & 1/s_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Homogenous form of rotation

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

Inverse rotation matrices,
$R^{-1}(\theta) = R(-\theta) = R^T(\theta)$

$$R(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}, \qquad R^T(\theta) = \begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Concatenation

- We perform 2 Translation on the same point

$$P' = P \cdot T(t_{x1}, t_{y1})$$
$$P'' = P' \cdot T(t_{x2}, t_{y2})$$
$$P'' = P \cdot T(t_{x1}, t_{y1}) \cdot T(t_{x2}, t_{y2}) = P \cdot T(t_{x1} + t_{x2}, t_{y1} + t_{y2})$$

So we expect:

$$T(t_{x1}, t_{y1}) \cdot T(t_{x2}, t_{y2}) = T(t_{x1} + t_{x2}, t_{y1} + t_{y2})$$

concatenation

The matrix product $T(t_{x1}, t_{y1}).\ T(t_{x2}, t_{y2})$ *is* :

$$\begin{bmatrix} 1 & 0 & t_{x1} \\ 0 & 1 & t_{y1} \\ 0 & 0 & 1 \end{bmatrix}. \begin{bmatrix} 1 & 0 & t_{x2} \\ 0 & 1 & t_{y2} \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_{x1} + t_{x2} \\ 0 & 1 & t_{y} + t_{y2} \\ 0 & 0 & 1 \end{bmatrix}$$

Properties of translations

1. $T(0,0) = I$
2. $T(t_{x1}, t_{y1}) \cdot T(t_{x2}, t_{y2}) = T(t_{x1} + t_{x2}, t_{y1} + t_{y2})$
3. $\mathrm{T}^{-1}(t_{x1}, t_{y1}) = T(-t_{x1}, -t_{y1})$

Concatenation of scales

The matrix product $S(s_{x1}, s_{y1}) \cdot S(s_{x2}, s_{y2})$ is :

$$\begin{bmatrix} s_{x1} & 0 & 0 \\ 0 & s_{y1} & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} s_{x2} & 0 & 0 \\ 0 & s_{y2} & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} s_{x1} \cdot s_{x2} & 0 & 0 \\ 0 & s_{y1} \cdot s_{y2} & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Only diagonal elements in the matrix – easy to multiply !

Properties of Scaling

1. $\mathrm{S}(1,1) = I$
2. $S(s_{x1}, s_{y1}) \cdot S(s_{x2}, s_{y2}) = S\left(s_{x1} \cdot s_{x2}, s_{y1} \cdot s_{y2}\right)$
3. $\mathrm{S}^{-1}(s_x, s_y) = S(1/s_x, 1/s_y)$

Concatenation of Rotation

$$\mathbf{R}(\theta)\cdot\mathbf{R}(\varphi)=\begin{bmatrix}\cos\theta & -\sin\theta & 0\\ \sin\theta & \cos\theta & 0\\ 0 & 0 & 1\end{bmatrix}\cdot\begin{bmatrix}\cos\varphi & -\sin\varphi & 0\\ \sin\varphi & \cos\varphi & 0\\ 0 & 0 & 1\end{bmatrix}=$$

$$\begin{bmatrix}\cos\theta\cos\varphi-\sin\theta\sin\varphi & -\cos\theta\sin\varphi-\sin\theta\cos\varphi & 0\\ \sin\theta\cos\varphi+\cos\theta\sin\varphi & -\sin\theta\sin\varphi+\cos\theta\cos\varphi & 0\\ 0 & 0 & 1\end{bmatrix}=\begin{bmatrix}\cos(\theta+\varphi) & -\sin(\theta+\varphi) & 0\\ \sin(\theta+\varphi) & \cos(\theta+\varphi) & 0\\ 0 & 0 & 1\end{bmatrix}=\mathbf{R}(\theta+\varphi)$$

Properties of rotation

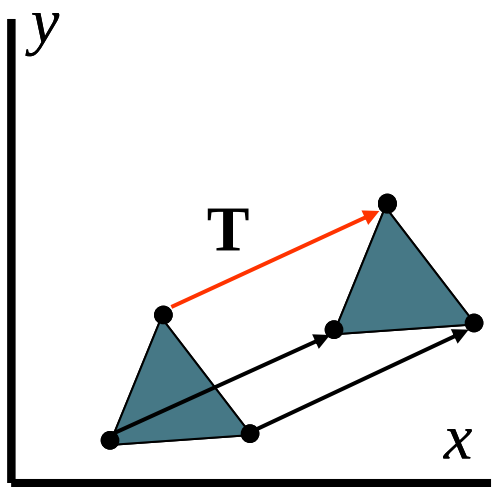
- $R(0) = I$
- $R(\theta) \cdot R(\varphi) = R(\theta + \varphi)$
- $R(\theta) \cdot R(\varphi) = R(\varphi) \cdot R(\theta)$
- Inverse rotation $R^{-1}(\theta) = R^{T}(\theta)$

2D polygon Transformation

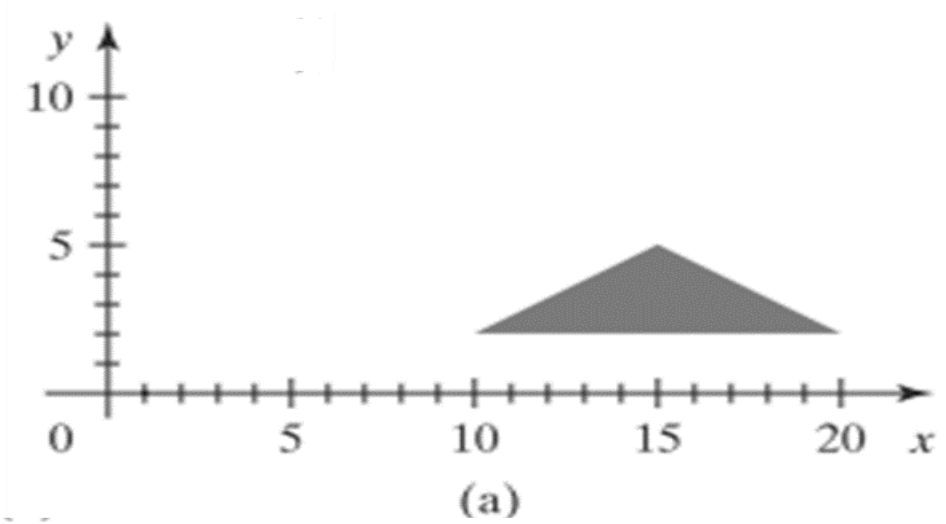
Translate polygon:

Apply the same operation on all points.

Works always, for all transformations of objects defined as a set of points.

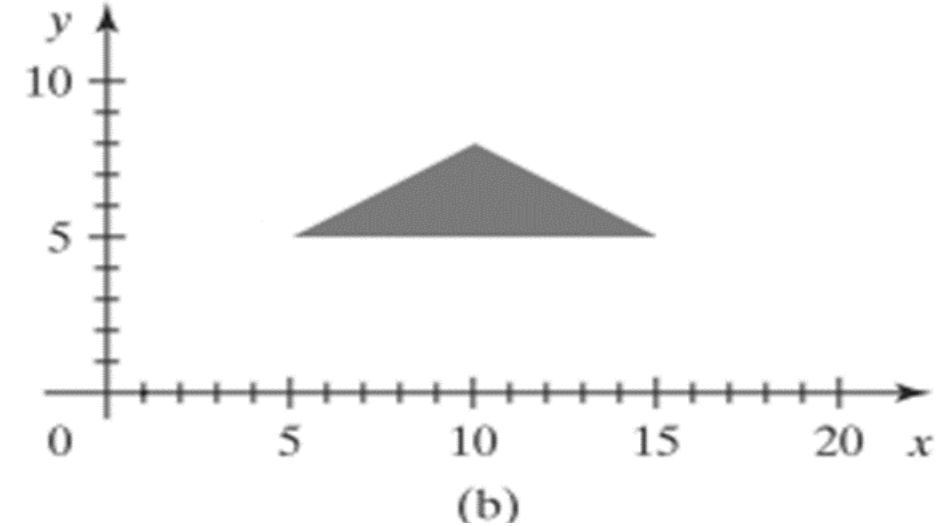


2D polygon Transformation



(a)

Moving a polygon from position (a) to position (b) with the translation vector (−5 , 3)



(b)

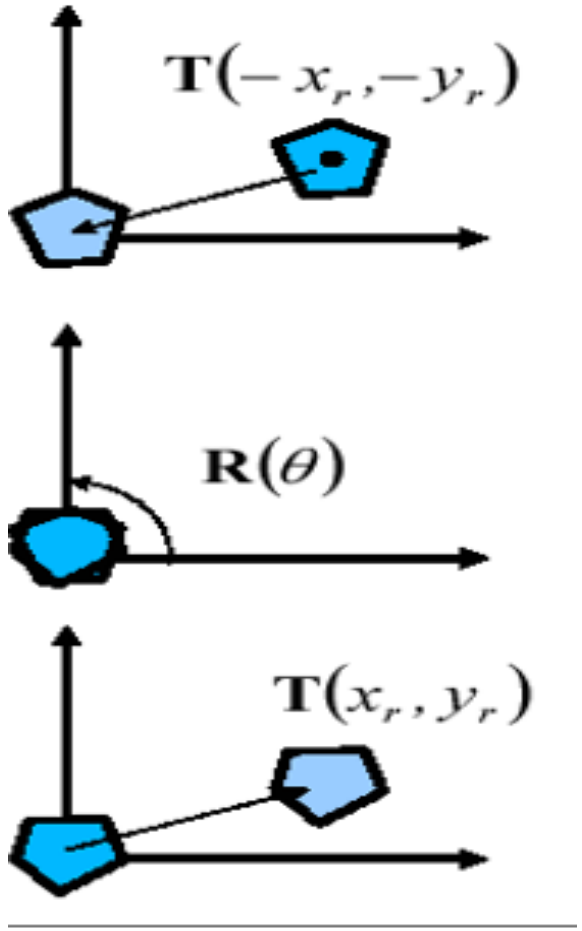
2D Composite Transformation

- Combine transformations of different type
  - translate, rotate, translate
  - translate, scale, translate
  - translate, reflect, translate
- Use to rotate or scale an object w.r.t. a point that is not the origin
- Implement by multiplication of the corresponding homogeneous matrices

Rotate an object w.r.t. a point that is not the origin

- Rotation around a pivot point
  1. Translate the object so that the pivot point moves to the origin
  2. Rotate around origin
  3. Translate the object so that the pivot point is back to its original position

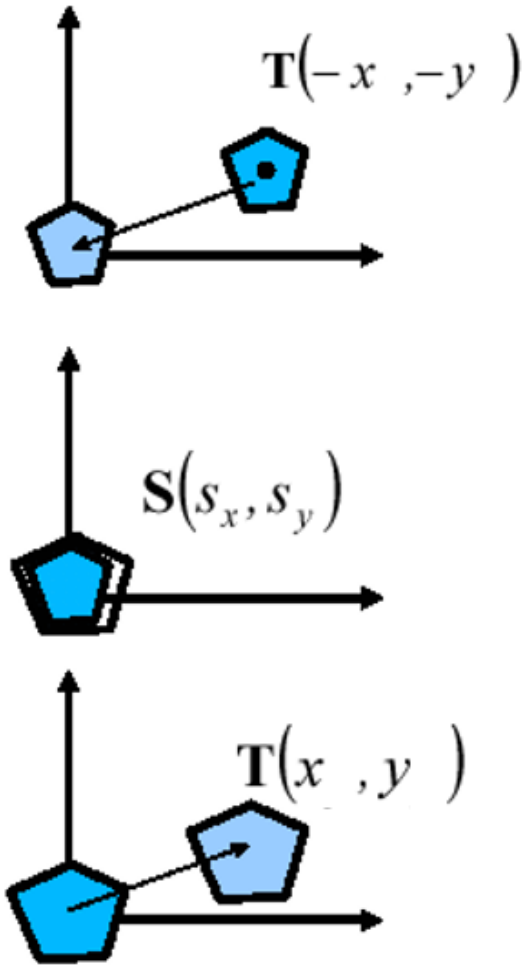
P'= $\mathrm{T}(t_x, t_y)$. **R(**θ**) .** $\mathrm{T}(-t_x, -t_y)$ **. p**



Scale an object w.r.t. a point that is not the origin

- Scale around a pivot point
  1. Translate the object so that the pivot point moves to the origin
  2. Scale around origin
  3. Translate the object so that the pivot point is back to its original position

$$\mathbf{P'=}\ T(t_x, t_y).\mathbf{S(}s_x, s_y\mathbf{)}\ .\ T(-t_x, -t_y)\ .\ \mathbf{p}$$



2D Composite Transformation

- SOLVE: Rotate Point p by angle 50 degree, then scale it to the double horizontally, then translate it 10 points to the bottom

  P'= R(50).P

  P'= S(2,1).R(50).p

  P'= T(0,-10). S(2,1).R(50).p

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -10 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} \cos 50 & -\sin 50 & 0 \\ \sin 50 & \cos 50 & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

2D Composite Transformation

- Rotate Point p by angle 40 degree around point(3,4)

P'= T(-3,-4).p
P'= R(40). T(-3,-4). P
P'= T(3,4). R(40). T(-3,-4). P

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 3 \\ 0 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} \cos 40 & -\sin 40 & 0 \\ \sin 40 & \cos 40 & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 0 & 0 & -3 \\ 0 & 0 & -4 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}$$

2D Composite Transformation

- Perform a Rotation by angle 90 degree of a triangle with vertices A(0,0), B(2, 1) and c(5, 4)
  - About origin
  - About the point (-2,-1)
  - About the triangle centroid

Answer:

About origin

P_A'= R(90). A→ $\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = . \begin{bmatrix} \cos 90 & -\sin 90 & 0 \\ \sin 90 & \cos 90 & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

P_B'= R(90). B → $\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = . \begin{bmatrix} \cos 90 & -\sin 90 & 0 \\ \sin 90 & \cos 90 & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$

P_C'= R(90). C → $\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = . \begin{bmatrix} \cos 90 & -\sin 90 & 0 \\ \sin 90 & \cos 90 & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 5 \\ 4 \\ 1 \end{bmatrix}$

2D Composite Transformation

- Perform a Rotation by angle 90 degree of a triangle with vertices A(0,0), B(2, 1) and c(5, 4)
  - About origin
  - About the point (-2,-1)
  - About the triangle centroid

Answer:

About point(-2,-1)

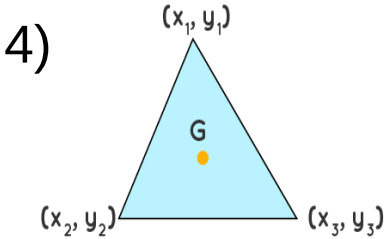
P_A'= T(-2,-1).R(90).T(2,1). A→
$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} \cos 90 & -\sin 90 & 0 \\ \sin 90 & \cos 90 & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

P_B'= T(-2,-1).R(90).T(2,1). B →
$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} \cos 90 & -\sin 90 & 0 \\ \sin 90 & \cos 90 & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$$

P_C'= T(-2,-1).R(90).T(2,1). C →→
$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -2 \\ 0 & 1 & -1 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} \cos 90 & -\sin 90 & 0 \\ \sin 90 & \cos 90 & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 5 \\ 4 \\ 1 \end{bmatrix}$$

2D Composite Transformation

Centroid Formula



$$G\left(\frac{x_1+x_2+x_3}{3}, \frac{y_1+y_2+y_3}{3}\right)$$

- Perform a Rotation by angle 90 degree of a triangle with vertices A(0,0), B(2, 1) and c(5, 4)
  - About origin
  - About the point (-2,-1)
  - About the triangle centroid

Answer:

- About the triangle centroid
- triangle centroid=((0+2+5)/3, (0+1+4)/3)=(2.33,1.66)

P_A'= T(2.33,1.66).R(90).T(-2.33,-1.66). A→

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 2.33 \\ 0 & 1 & 1.66 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} \cos 90 & -\sin 90 & 0 \\ \sin 90 & \cos 90 & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 1 & 0 & -2.33 \\ 0 & 1 & -1.66 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

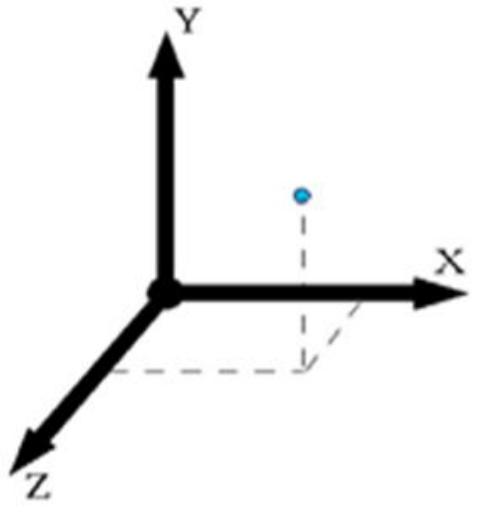
P_B '= T(2.33,1.66).R(90).T(-2.33,-1.66). B →

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 2.33 \\ 0 & 1 & 1.66 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} \cos 90 & -\sin 90 & 0 \\ \sin 90 & \cos 90 & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 1 & 0 & -2.33 \\ 0 & 1 & -1.66 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$$

P_C'= T(2.33,1.66).R(90).T(-2.33,-1.66). C →

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 2.33 \\ 0 & 1 & 1.66 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} \cos 90 & -\sin 90 & 0 \\ \sin 90 & \cos 90 & 0 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 1 & 0 & -2.33 \\ 0 & 1 & -1.66 \\ 0 & 0 & 1 \end{bmatrix} . \begin{bmatrix} 5 \\ 4 \\ 1 \end{bmatrix}$$

Transformation in 3D



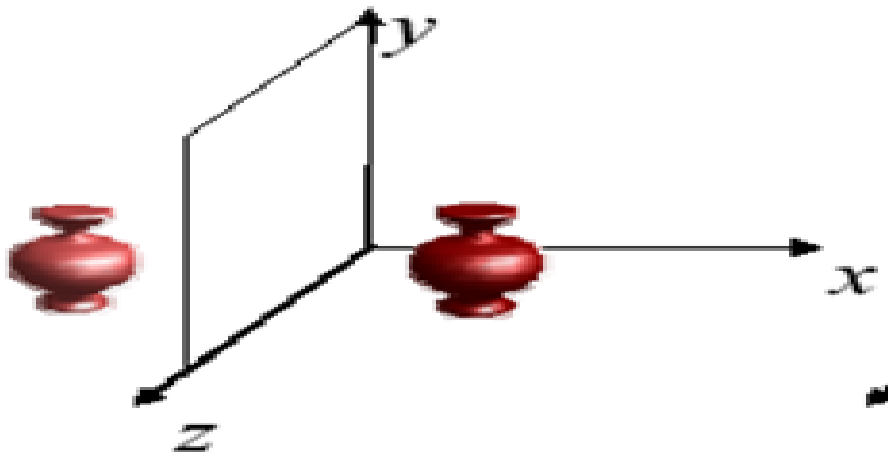
- How should we treat geometric transformations in 3D?
- Just add one more coordinate/axis!
- A point is represented as $\begin{bmatrix} x \\ y \\ z \end{bmatrix}$
- But remember to use homogeneous coordinates! Embed scale and rotation matrices as upper left sub matrices and translation vectors as upper right sub vectors of the right column $\begin{bmatrix} x \\ y \\ z \\ 1 \end{bmatrix}$

Transformation in 3D

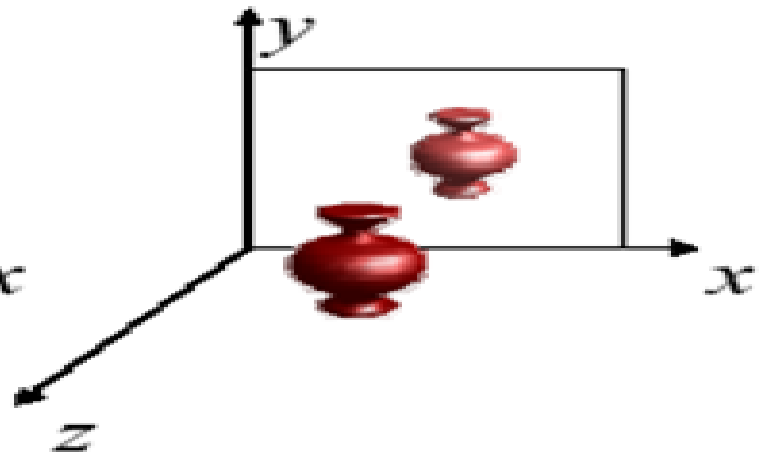
Transformation	Matrix	Comments
Scaling	$\begin{bmatrix} s_x & 0 & 0 & 0 \\ 0 & s_y & 0 & 0 \\ 0 & 0 & s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$	Looks just like the 2D version. We just added an s_z term.
Rotation	(see next slide)	In 2D, only one axis of rotation; now there are infinitely many! Must take all into account. See next slide...
Translation	$\begin{bmatrix} 1 & 0 & 0 & dx \\ 0 & 1 & 0 & dy \\ 0 & 0 & 1 & dz \\ 0 & 0 & 0 & 1 \end{bmatrix}$	Similar to the 2D version, just with one more entry dz, representing change in the z-direction.

Reflection in 3D

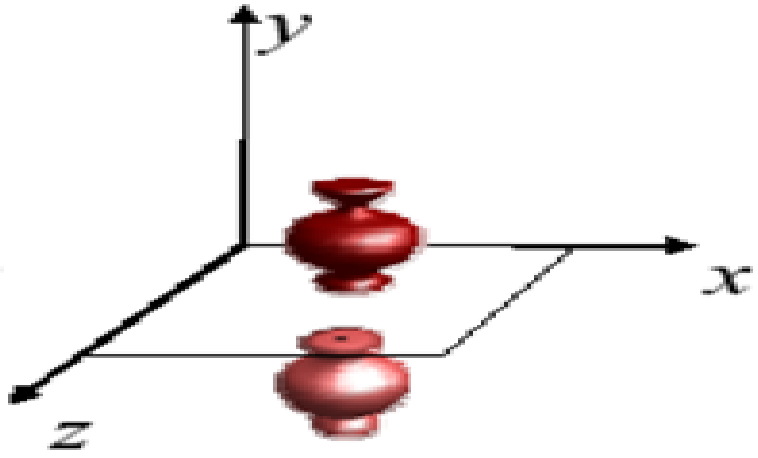
- Reflection over planes



$$\begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



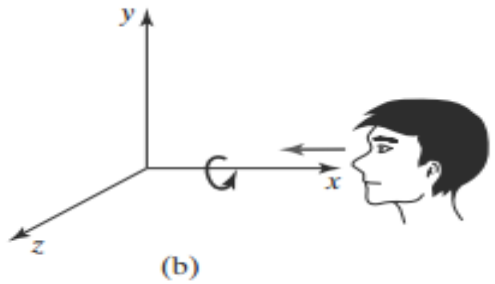
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$



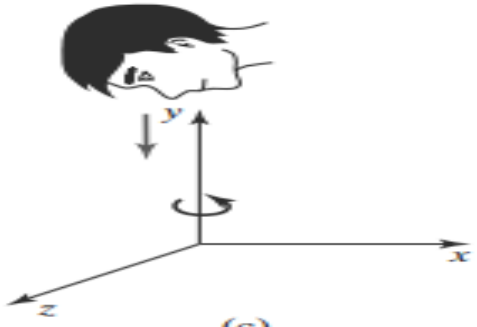
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Rotation in 3D

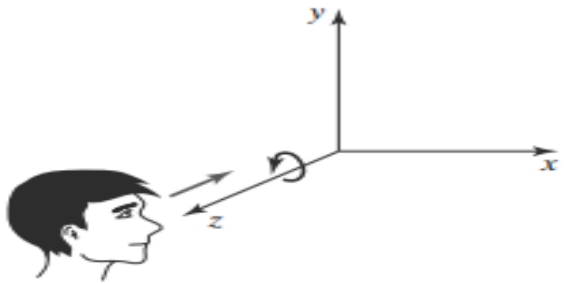
- Need to specify which axis the rotation is about.
- Counterclockwise when looking along the positive half towards the origin.



(b)



(c)



(a)

x axis

$$\mathbf{R}_x(\theta) = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta & 0 \\ 0 & \sin\theta & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

y axis

$$\mathbf{R}_y(\theta) = \begin{bmatrix} \cos\theta & 0 & \sin\theta & 0 \\ 0 & 1 & 0 & 0 \\ -\sin\theta & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

z axis

$$\mathbf{R}_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 & 0 \\ \sin\theta & \cos\theta & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Inverses and composition in 3D

Transformation	Inverse Matrix
Scaling	$\begin{bmatrix} 1/s_x & 0 & 0 & 0 \\ 0 & 1/s_y & 0 & 0 \\ 0 & 0 & 1/s_z & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Rotation	$\boldsymbol{R_{yz}^{-1}(\psi)}$ $\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \cos\psi & \sin\psi & 0 \\ 0 & -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$, $\boldsymbol{R_{zx}^{-1}(\theta)}$ $\begin{bmatrix} \cos\theta & 0 & -\sin\theta & 0 \\ 0 & 1 & 0 & 0 \\ \sin\theta & 0 & \cos\theta & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$, $\boldsymbol{R_{xy}^{-1}(\phi)}$ $\begin{bmatrix} \cos\varphi & \sin\varphi & 0 & 0 \\ -\sin\varphi & \cos\varphi & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$
Translation	$\begin{bmatrix} 1 & 0 & 0 & -dx \\ 0 & 1 & 0 & -dy \\ 0 & 0 & 1 & -dz \\ 0 & 0 & 0 & 1 \end{bmatrix}$

- Composition works exactly the same way...

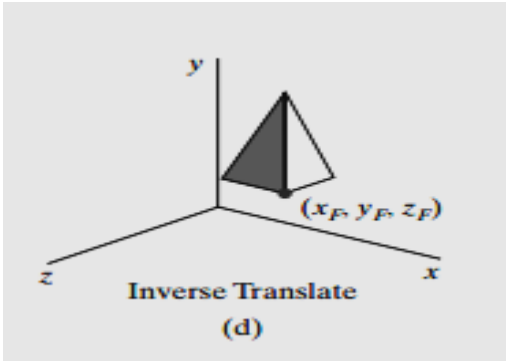
Scale an object w.r.t. a point that is not the origin in 3D

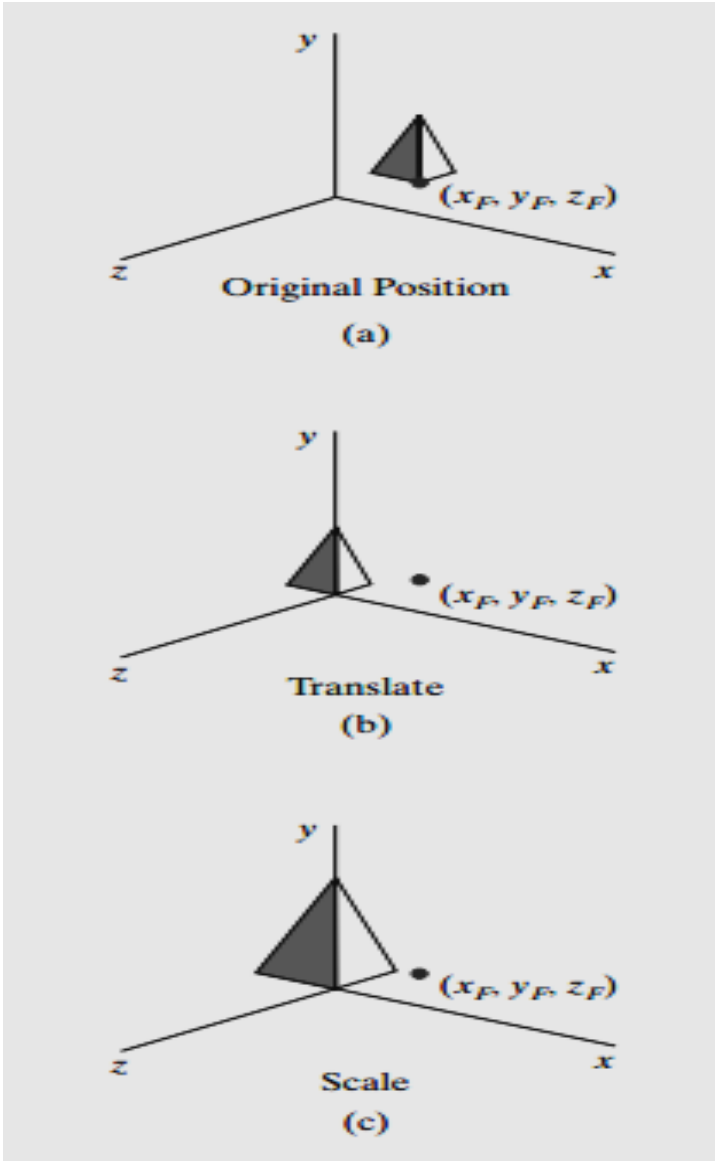
some graphics packages provide only a routine that scales relative to the coordinate origin, we can always construct a scaling transformation with respect to any selected *fixed position(x f , y f , z f)* using the following transformation sequence:

1. Translate the fixed point to the origin.

2. Apply the scaling transformation relative to the coordinate origin

3. Translate the fixed point back to its original position





Thank You